

Total No. of Questions : 8]

PB-2373

SEAT No. :

[Total No. of Pages : 2

[6263]-223

B.E. (Mechanical Engineering)
Computer Integrated Manufacturing
(Semester-VIII) (2019 Pattern) (402048)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q. 7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of electronic pocket calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Explain with neat sketch any three canned cycles with examples. [9]

b) Write a CNC turning program for following component shown in Figure 1b. Assume suitable data. [9]

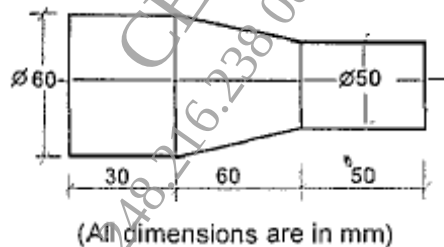


Fig. 1b

OR

Q2) a) Explain G00, G01, G02 code in part programming with suitable example. [9]

b) Write a complete part program for milling machine on a work piece shown in Figure 2b. Assume suitable speed and feed for machining. [9]

P.T.O.

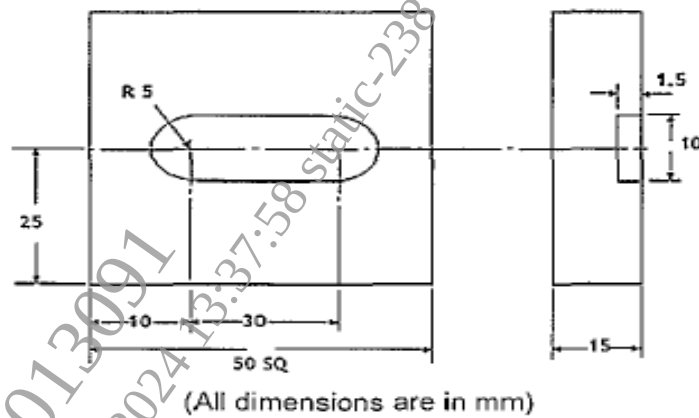


Fig. 2b

- Q3)** a) Explain Computer Aided Process Planning (CAPP) and its Benefits. [9]
 b) Explain in brief [8]
 i) Capacity planning,
 ii) Enterprise Resource Planning (ERP).

OR

- Q4)** a) Describes the different approaches to Computer Aided Process Planning (CAPP). [9]
 b) Describe the Manufacturing Resource Planning (MRP-II) with input, working, outputs and benefits. [8]

- Q5)** a) Explain cellular manufacturing and types of machine cell design with layouts. [9]
 b) Describe Rank Order Clustering (ROC) algorithm with suitable example. [9]

OR

- Q6)** a) Explain Optiz parts coding system with suitable example. [9]
 b) Explain any three Flexible Manufacturing Systems (FMS) based on layout with suitable sketches. [9]
Q7) a) State and explain key IoT Applications in Manufacturing. [9]
 b) Explain Big-Data and Cloud Computing used for IoT along with its benefits and limitations. [8]

OR

- Q8)** a) Explain the use of IoT for Smart Manufacturing, Predictive Maintenance and Supply-Chain & Logistics applications. [9]
 b) Explain with an example, How Digital Twin is implemented for Smart Manufacturing? [8]

